Navaneeth Kombilath

Kannur,Kerala navaneeth123k@gmail.com (+91) 8156868078 navaneeth-k

Education

Bachelors-Masters Dual Degree (BS-MS) *IISER Thiruvananthapuram*
5th year Physics Major Student Aug 2021 – Current
Minor in Mathematics

- Current CGPA: **7.48/10**

Higher Secondary School Record *Kendriya Vidyalaya Kannur*
All India Senior School Certificate Exam (AISSCE) *Apr 2019 – Mar 2020*

- Percentage: 95%

Research Experience

- **Dark Matter Constraints With Neutron Star Cooling – Master's thesis** **LPTHE & IISER TVM**
  Supervisors: Prof. Silvia Manconi (LPTHE, Sorbonne Université) *April 2025 – Present*
  Prof. Mathew Arun Thomas (IISER TVM)
  Consultation: Prof. Dany Page (UNAM) – NSCool implementation
  - **Phase I (IISER TVM):** Independently solved the Tolman–Oppenheimer–Volkoff equations for several realistic nuclear equations of state to obtain the structural profiles of dark-matter–admixed neutron stars. Report
  - **Phase II (LPTHE):** Re-derived the neutrino emissivity associated with DM-induced heating from the Cermeño et al. interaction Lagrangian at amplitude level, and extended the calculation to DM-induced neutron decay – an emissivity channel for which no prior expression was available in the literature. Report
  - Implemented both heating mechanisms as a dedicated subroutine inside NSCool in consultation with Prof. Dany Page; the resulting cooling curves exhibit clear deviations from standard neutron-star evolution.

- **Nucleon - Light Dark Matter Annihilation through Baryon Number Violation** **IISER TVM**
  Guide: Dr. Mathew Arun Thomas *Nov 21, 2024 – Jan 21, 2025*
  – Investigated baryon-number-violating dark-matter annihilation in colour-triplet extensions of the Standard Model, focusing on fermionic dark-matter stability near the proton mass.
  – Derived constraints on the mediator scale (m_Φ) up to 10^7 GeV from Super-Kamiokande data, and compared the resulting Majorana and Dirac dark-matter bounds.

- **Modeling Dark Matter Freeze-Out Using the Boltzmann Equation** **IIT Guwahati**
  (Summer Project) *May 21, 2024 – June 28,2024*
  Guide: Dr. Subhaditya Bhattacharya
  - Developed a numerical model of the Boltzmann equation to analyze the thermal freeze-out mechanism and understand how dark matter decoupled from the early universe's thermal bath.
  - Utilized Python and Mathematica to solve differential equations and visualize the evolution of dark matter density, comparing results with observational data. Report

- **Six-Dimensional Spacetime with (3,3) Signature: Quantum Nonlocality and Spacetime Symmetry TIFR, Pune**
  Guide: Prof. Tejinder Pal Singh *May 07, 2024- Nov 28, 2024*
  – Worked within Prof. Singh's six-dimensional (3,3)-signature spacetime framework, exploring how classical and quantum sectors can be embedded in different submanifolds and how weak-interaction structure admits a spacetime-symmetry interpretation.
  – The project strengthened my background in abstract field-theoretic model building. Report

- **Analysis of the duration–hardness ratio plane of GRBs**
  (Self-study project) **IIT Hyderabad**
  Guide: Prof. Shantanu Desai *May 27, 2023 – Aug 15, 2023*
  –Gained a basic understanding of data analysis methods under the guidance of Prof. Shantanu Desai by reproducing results from the research article titled "Analysis of the duration–hardness ratio plane of gamma-ray bursts using skewed distributions."

Academic achievements

- **Microfluidic Device Development**: Developed a prototype microfluidic device for diagnosing clinical depression as part of an international competition held in Paris, securing the **Best Diagnostics Project** title and a **Gold Medal**. Awards
- **Indian Patent Granted (No. 569461, 2024)**: Co-inventor on the patent "A Photodarlington Sensor-Based Device for Fluorometric Analysis and a Process Thereof," granted by the Government of India for hardware developed during the iGEM 2023 OASYS project. Patent
- **Academic Achievement**: Ranked in the **top 3%** of students in the All-India Senior School Certificate Examination among 1.4 million students from Kendriya Vidyalaya schools across India.

Posters and Conference

- **"Neutron Star Cooling due to Dark Matter–Induced Neutron Decay"**, Poster, National Undergraduate Physics Conference (NUPC), IISER Tirupati, **December 2025**.
- **Frontier Symposium in Physics 2024 (FSP 2024)**, Participated in the conference at IISER Thiruvananthapuram, 2024

Skill Sets

- **Scientific computing**: Python (NumPy, SciPy, Matplotlib), Mathematica, Fortran (NSCool modifications); numerical integration of stiff ODE systems including the Tolman–Oppenheimer–Volkoff equations.
- **Physics methods**: amplitude-level calculations from BSM interaction Lagrangians; Boltzmann-equation integration for relic-abundance studies; structural and thermal modelling of compact objects; nuclear equations of state.
- **Tools**: LaTeX, Git.
- **Coursework**: Quantum Field Theory; Quantum Information Theory; General Relativity and Cosmology; High Energy & Particle Physics; Condensed Matter Physics (I & II); Foundations of Quantum Mechanics; Advanced Electrodynamics; Quantum Mechanics II; three Advanced Laboratory Courses on Optics, Electronics and Mechanics.
- **Languages**: English, Hindi, Malayalam, Tamil.

Leadership and Management

Core Committee member of ESI (Ecological Club of IISER TVM) *2022-2024*
Conducted Nature Walks, Bird Watching and events for Anvesha (Science fest of IISER TVM)
Managed the Social media handle for the Club

Hardware Team Head - iGEM 2023 *2022- 2023*
Build a hardware device - OASYS as a part of the iGEM project and lead the hardware team to Final Grand Jamboree at Paris, France

Hospitality Head for ISHYA (Cultural fest of IISER TVM) *2023-2024*
Core Committee member Hospitality Head for two consecutive years for the event and conducted the event smoothly.